7.3 Basic concepts of immunology

Immunology is the study of immune system. This system protects an indvidual from various infective agents. It refers to all the mechanisms used by the body for protection from environmental agents that are foreign to the body.

When the immune system does not function efficiently in an individual, it leads to infection causing disease. The overall ability of body to fight against the disease causing pathogen is called **immunity**. It is also called disease resistance and the lack of immunity is known as susceptibility. Immunity is highly specific.

Normally many of the responses of the immune system initiate the destruction and elimination of invading organisms and any toxic molecules produced by them. These immune reactions are destructive in nature and are made in response only to molecules that are foreign to the host and not to those of host itself. This ability to distinguish foreign molecules from self is another fundamental feature of the immune system. However, occasionally, it fails to make its distinction and reacts destructively against the host's own molecules; such autoimmune diseases can be fatal to the organism.

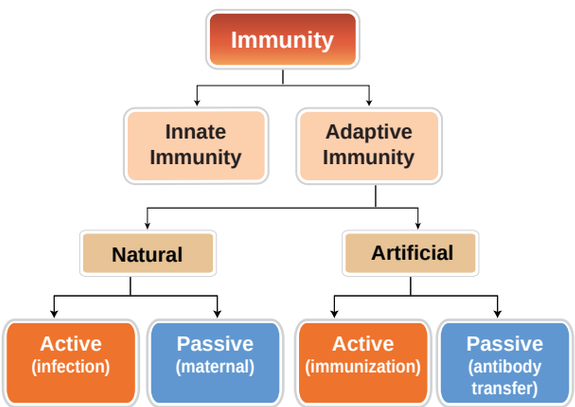


Fig. 7.9 Immune system

Almost all the macromolecules e.g. proteins, polysaccharides, nucleic acids, etc., as long as they are foreign to recipient organism can induce immune response. Any substance capable of eliciting immune response is called an **ANTIGEN** (**ANTI**body **GEN**erator). There are two broad classes of immunity responses namely, innate immunity and acquired immunity **(Fig. 7.9)**.

7.3.1 Innate immunity

Innate immunity is the natural phenomenon of resistance to infection which an individual possesses right from the birth. The innate defense mechanisms are non-specific in the sense that they are effective against a wide range of potentially infectious agents. It is otherwise known as **non-specific immunity** or **natural immunity.**

A number of innate defense mechanisms are operative non-specifically against a large number of microorganisms as shown in the **Table 7.4** and **Fig. 7.10.**

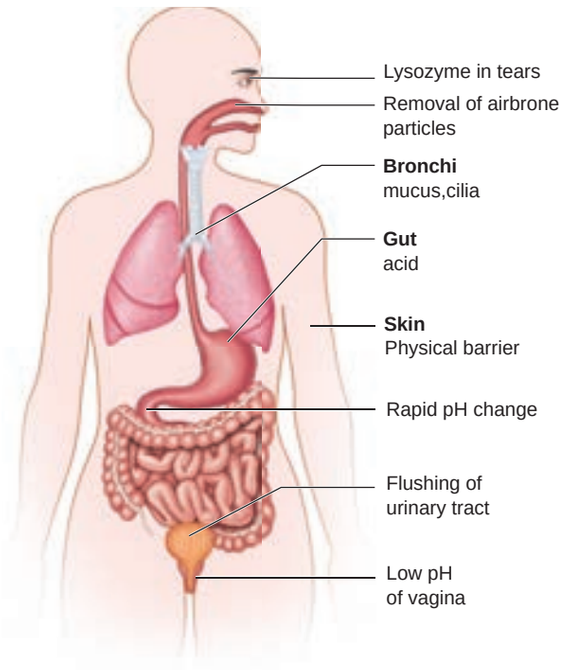


Fig. 7.10 Various anatomical and physiological barriers to microbial attack

TN_GOVT_XII_Bio_Zoology_EM_chapter 7.indd 116 20-03-2020 10.34.19 AM